

The Derivative of a Function

Section 2.2

A. Definition of the Derivative

For a function $f(x)$ the derivative is $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Examples: Using the definition of the derivative, find the derivative of the following functions:

1. (video) $f(x) = 2x^2 + 8x - 2$

2. (video) $g(x) = 2\sqrt{x} + 9$

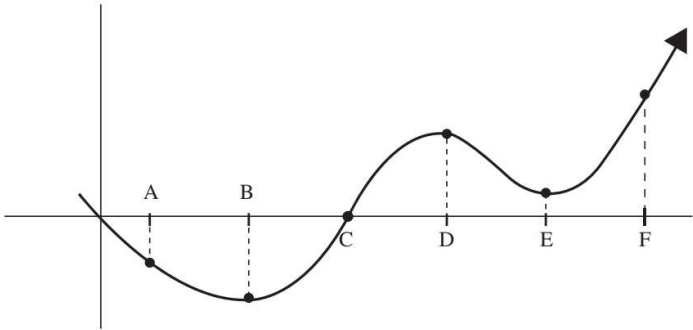
3. (video) $h(x) = \frac{5}{3x+1}$

4. $f(x) = x^2 + 3x - 2$

5. $k(x) = \frac{2}{x}$

6. $g(x) = \sqrt{2x+1}$

7. Consider the graph for the function $f(x)$



Estimate the following:

- a. $f'(A)=$

b. $f'(B)=$
- c. $f'(C)=$

d. $f'(D)=$
- e. $f'(E)=$

f. $f'(F)=$

B. Notation

Function	Derivative
$y =$	$y' =$ $\frac{dy}{dx} =$ (Leibniz notation)
$f(x)$	$f'(x)$
$F(x)$	$f(x)$

C. Differentiability

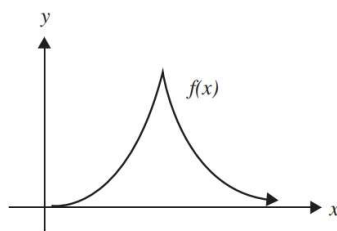
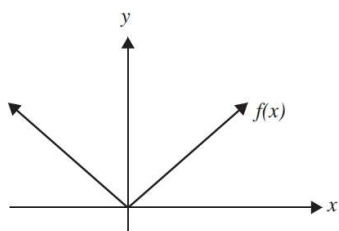
DEFINITION: We say that

- $f(x)$ is differentiable at a if $f'(a)$ exists
- $f(x)$ is differentiable on (a, b) if it is differentiable on every point in (a, b)

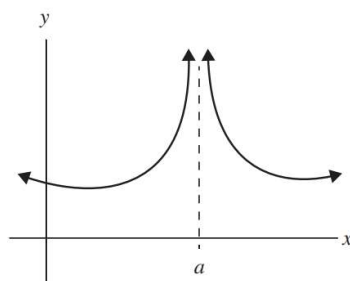
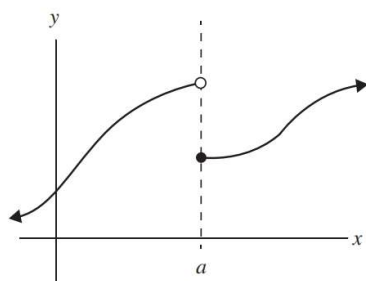
Theorem: If f is differentiable at a , then f is continuous at a

Video Nondifferentiable functions

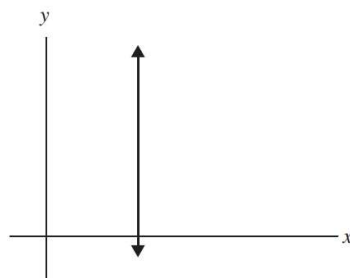
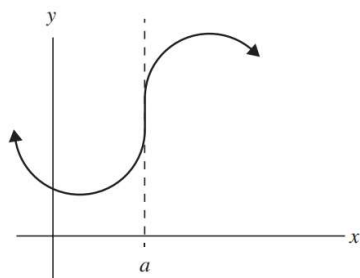
i. Any function with a “corner” or cusp



ii. Any function with a discontinuity

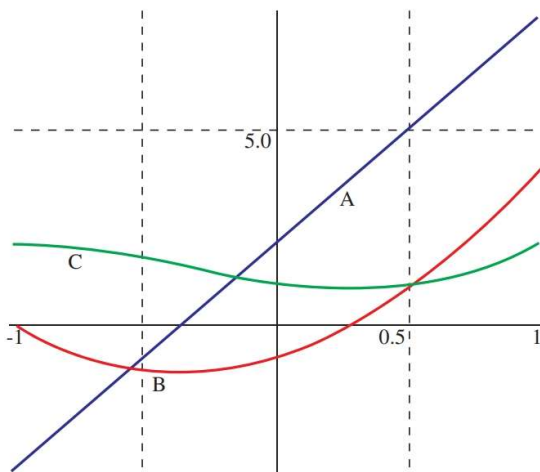


iii. Any function with a vertical tangent



More Examples

8.



Identify the graphs A (blue), B (red), and C (green) as the graphs of a function and its derivatives.

a. The graph of the function is:

b. The graph of the function's first derivative is:

c. The graph of the function's second derivative is:

9. Evaluate the limit $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$

10. Evaluate the limit $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 36} - 6}{x^2}$.